MIL-F-47254(MI)
26 July 1974
SUPERSEDING
MIS-13912
17 June 1966

MILITARY SPECIFICATION

FOAM, POLYURETHANE, OPEN CELL, MEDIUM FLEXIBILITY

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of open cell polyurethane foam, prefoamed, of medium flexibility.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordnance
MIL-C-26861	Cushioning Material, Resilient Type, General

STANDARDS

Military

MIL-STD-105	Sampling, Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM D 634	Methods of Test for Product Uniformity of Phenolic Laminated Sheets
ASTM D 1564	Tentative Specifications and Methods of Test for Flexible Urethane Foam

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19105.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), a preproduction sample shall meet all the requirements of this specification.

3.2 Materials.

3.2.1 Composition and physical properties.

3.2.1.1 Composition. The materials used in the manufacture of the polyurethane foam shall be of high quality and entirely suitable for the purpose intended and shall provide a product conforming to the requirements specified herein.

3.2.1.2 Form. The polyurethane foam shall be prefoamed and supplied in rolls, flat sheet, or molded shapes (see 6.2c).

3.2.2 Dimensions.

3.2.2.1 Length and width. The tolerance on the length of rolls shall be within minus 1 to plus 5 percent of the specified length (see 6.2). The tolerance on the width of a roll and the length and width of a sheet shall

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be minus 1 to plus 10 percent of the specified dimensions (see 6.2). For molded shapes, the tolerances shall be in accordance with good practice commensurate with the application or as specified by the procuring activity (see 6.2).

3.2.2.2 Thickness. The tolerance on thickness (see 6.2) shall be minus 10 to plus 20 percent of the specified thickness or plus or minus 1/8 inch, whichever is the smaller.

3.2.3 Density. The density shall be 4.0 plus or minus 0.5 pounds per cubic foot.

3.2.4 Fusion. The foam shall show no tendency to fuse when die cut.

3.3 Performance characteristics.

3.3.1 Resiliency. The time to recover 95 percent of foam's original thickness shall not exceed 5 seconds.

3.3.2 Compression set. Permanent set from compression shall not exceed 10 percent.

3.3.3 Indentation. The load required to compress the foam to 75 percent of its thickness shall be 55 pounds minimum and 95 pounds maximum.

3.4 Environmental conditions.

3.4.1 Flame resistance. The foam shall be self-extinguishing in 2 seconds maximum.

3.4.2 Low temperature recovery. After exposure to minus 40 degrees plus or minus 2 degrees Fahrenheit (F) (minus 40 degrees plus or minus 1 degree Celsius (C)) for at least 5 hours, the foam shall return to 80 percent of its original thickness.

3.4.3 Dry heat resistance. After 48 hours exposure to 212 degrees plus or minus 5 degrees F (100 degrees plus or minus 3 degrees C) the foam shall return to 88 percent of its original thickness.

3.4.4 High temperature resistance. The foam shall exhibit no loss of tensile strength and elongation after 168 hours at 160 degrees plus or minus 5 degrees F (71 degrees plus or minus 3 degrees C).

3.4.5 Resistance to hydraulic fluid. There shall be less than 5 percent change in any mechanical property after immersion in hydraulic fluid conforming to MIL-H-5606 for 7 days.

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3.5 Workmanship. The polyurethane foam shall be uniform in quality, and shall be free from voids, cracks, impurities and other defects that could adversely affect its use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and service conform to prescribed requirements.

4.2 Preproduction inspection. When specified in the contractual document (see 6.2), a sample consisting of 1 piece 4 inches by 24 inches by 48 inches, manufactured under the same conditions as those proposed for subsequent production, shall be subjected to preproduction inspection. Accomplishment of preproduction inspection shall be as specified herein. If the preproduction sample does not meet the requirements of this specification, it shall be rejected. Subsequent units shall not be considered for acceptance until Government approval of the preproduction sample has been obtained. Units subjected to preproduction inspection shall have successfully passed acceptance inspection. Testing of the preproduction sample, to determine compliance with the characteristics listed in Table I, shall be conducted in accordance with the corresponding test paragraphs.

Table I. Preproduction inspection

Characteristic	Requirement source	Test paragraph
Composition	3.2.1.1	4.5.4.1
Density	3.2.3	4.5.4.3
Fusion	3.2.4	4.5.4.4
Resiliency	3.3.1	4.5.4.5
Compression set	3.3.2	4.5.4.6
Indentation	3.3.3	4.5.4.7
Flame resistance	3.4.1	4.5.4.8
Low temperature recovery	3.4.2	4.5.4.9
Dry heat resistance	3.4.3	4.5.4.10
High temperature resistance	3.4.4	4.5.4.11
Resistance to hydraulic fluid	3.4.5	4.5.4.12
Workmanship	3.5	4.5.5

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4.3 Quality conformance inspection.

4.3.1 Sampling for acceptance inspection. Sampling for acceptance inspection shall be conducted in accordance with MIL-STD-105, at an acceptable quality level (AQL) of 4.0 percent defective.

4.3.2 Lot formation. A lot shall consist of all the material manufactured in one continuous operation by the same process by the same manufacturer in accordance with this specification and submitted for inspection at one time.

4.3.3 Acceptance inspection. Acceptance inspection of the sample specified in 4.3.1, to determine compliance with the characteristics specified in Table II, shall be conducted in accordance with the corresponding test and inspection paragraphs.

Table II. Acceptance inspection

Characteristic	Requirement source	Test and inspection paragraph
Form	3.2.1.2	4.5.4.1
Length and width	3.2.2.1	4.5.4.2
Thickness	3.2.2.2	4.5.4.2
Density	3.2.3	4.5.4.3
Resiliency	3.3.1	4.5.4.5
Compression set	3.3.2	4.5.4.6
Workmanship	3.5	4.5.5

4.4 Inspection equipment. The inspection equipment for conducting examination and tests shall be as specified in the applicable test methods and procedures paragraphs.

4.5 Test methods and procedures.

4.5.1 Test conditions. Unless otherwise specified herein, the following conditions shall be used as a basis to establish performance requirements.

a. Temperature, room ambient plus or minus 9 degrees C (16 degrees F).

b. Altitude, facility ground.

c. Humidity, facility ambient up to 95 percent relative humidity.

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4.5.2 Test sequence. Testing within each classification of inspection (preproduction or acceptance) shall be at the option of the supplier.

4.5.3 Test specimen configuration. Unless otherwise specified herein all specimens shall be a rectangular block having dimensions of 8 inches by 8 inches by 4 inches.

4.5.4 Tests.

4.5.4.1 Composition and physical properties. Composition shall be certified by the supplier and form shall be examined visually to establish compliance to 3.2.1.

4.5.4.2 Dimensions. Dimensional tolerance shall be meansured to establish compliance to 3.2.2.

4.5.4.3 Density. The density shall be determined in accordance with MIL-C-26861 and shall conform to 3.2.3.

4.5.4.4 Fusion. The specimen shall be die cut and visually examined for conformance to 3.2.4.

4.5.4.5 Resiliency. The specimens shall be compressed to 25 percent of the original thickness and held for 1 minute, and the load shall then be removed. The specimens shall be allowed to recover under no load. The foam shall conform to 3.3.1.

4.5.4.6 Compression set. Compression set shall be determined in accordance with ASTM D 1564, Method B, and shall conform to 3.3.2.

4.5.4.7 Indentation. Indentation shall be determined in accordance with ASTM D 1564, Method A, and shall conform to 3.3.3.

4.5.4.8 Flame resistance. Flame resistance shall be determined in accordance with ASTM D 634 except that the time of exposure to flame shall be 15 seconds plus or minus 1 second. The time for extinguishing shall be counted from the instant of removal of burner. The foam shall conform to 3.4.1

4.5.4.9 Low temperature recovery. The low termperature recovery shall be tested in accordance with ASTM D 1055 and shall conform to 3.4.2.

4.5.4.10 Dry heat resistance. A 15 by 15 inch specimen, 4 inches thick, shall be heated in a recirculating air oven at 212 degrees plus or minus 5 degrees F (100 degrees plus or minus 3 degrees C) for at least 48 hours. The specimens shall be tested for indentation as specified in ASTM D 1564 and shall conform to 3.4.3.

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4.5.4.11 High temperature resistance. The tensile specimens, of the size specified in MIL-C-26861, shall be placed at 160 degrees plus or minus 5 degrees F (71 degrees plus or minus 3 degrees C) for at least 168 hours. The tensile test shall be conducted at room temperature in accordance with MIL-C-26861. The foam shall conform to 3.4.4.

4.5.4.12 Resistance to hydraulic fluid. The test specimens shall be immersed in MIL-H-5606 hydraulic fluid for 7 days. The specimens shall then be removed, squeezed and wiped dry, then tested in accordance with 4.5.4.11, and shall conform to 3.4.5.

4.5.5 Workmanship. The foam shall be visually examined to assure conformance to the workmanship requirement specified in 3.5.

4.6 Packaging, packing, and marking. The inspector shall ascertain that the packaging, packing, and marking of the material conforms to this specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. Unless otherwise specified by the procuring activity, the packaging and packing shall be in accordance with the best accepted commercial practice.

5.2 Marking. Marking shall be in accordance with MIL-STD-129. Marking shall include, but not be limited to the following information:

a. Manufacturer's name and address.

b. Manufacturer's designation.

c. Lot number, or batch number.

d. Date of manufacture.

e. Number of this specification.

6. NOTES

6.1 Intended use. The material is intended for use in packaging applications.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

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b. Whether a preproduction sample is required (see 3.1), and if so, pertinent details.

c. Form required (see 3.2.1.2).

d. Dimensions required (see 3.2.2.1 and 3.2.2.2).

e. Requirement for certification of composition (see 4.5.4.1).

f. Color, if different (see 6.3).

6.3 Color. Unless otherwise specified (see 6.2), the foam shall be charcoal in color.

6.4 Supersession data. This specification includes the requirements of Missile Interim Specification MIS-13912, dated, 17 June 1966.

Custodian:
ARMY-MI

Preparing Activity:
ARMY - MI
Project No. 9330-A599

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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